Activity 2.3 – Synchronous FIFO

1.Design Approach:

Designed a parameterized synchronous FIFO (First In First Out) memory using Verilog HDL. Implemented memory storage, read and write pointers, and full and empty status flags.

2.Verification Steps:

- Created a separate testbench.
- Applied reset to initialize FIFO pointers.
- Tested write operations.
- Tested read operations.
- Verified data ordering.
- Checked full and empty flags.
- Simulated using ModelSim.

3.Results:

The testbench checks FIFO data storage, retrieval and status flags. PASS/FAIL messages are displayed.

4.Issues Faced:

Read and write pointers must be updated correctly. FIFO overflow and underflow conditions must be avoided. Full and empty flags require accurate counter management.

5.Conclusion:

Successfully designed and verified a synchronous FIFO memory using Verilog HDL.